Module 5 — Physical Chemistry and Transition Elements

5.1 How Fast? — Rate Equations

Rate Equations and Orders

Rate equation: An experimentally determined expression that relates the rate of a reaction to the concentrations of reactants raised to certain powers (orders). For a reaction A + B → products: **Rate = $k[A]^m[B]^n$**, where m and n are the orders with respect to A and B, and k is the rate constant. The rate equation can only be determined from experimental data, *not* from the balanced equation.

Order of reaction: The power to which the concentration of a reactant is raised in the rate equation. The **overall order** is the sum of the individual orders (m + n). Orders are always small whole numbers (0, 1, or 2) at A-level.

- **Zero order:** Changing the concentration has no effect on rate. Rate vs [A] graph is a horizontal line. Concentration-time graph is a straight line with negative gradient.
- **First order:** Rate is directly proportional to concentration. Rate vs [A] gives a straight line through the origin. The concentration-time graph shows **exponential decay with a constant half-life** — this is the key test for first order.
- **Second order:** Rate is proportional to concentration squared. Rate vs [A] gives an upward curve; Rate vs $[A]^2$ gives a straight line through the origin. Half-life increases as reaction proceeds.

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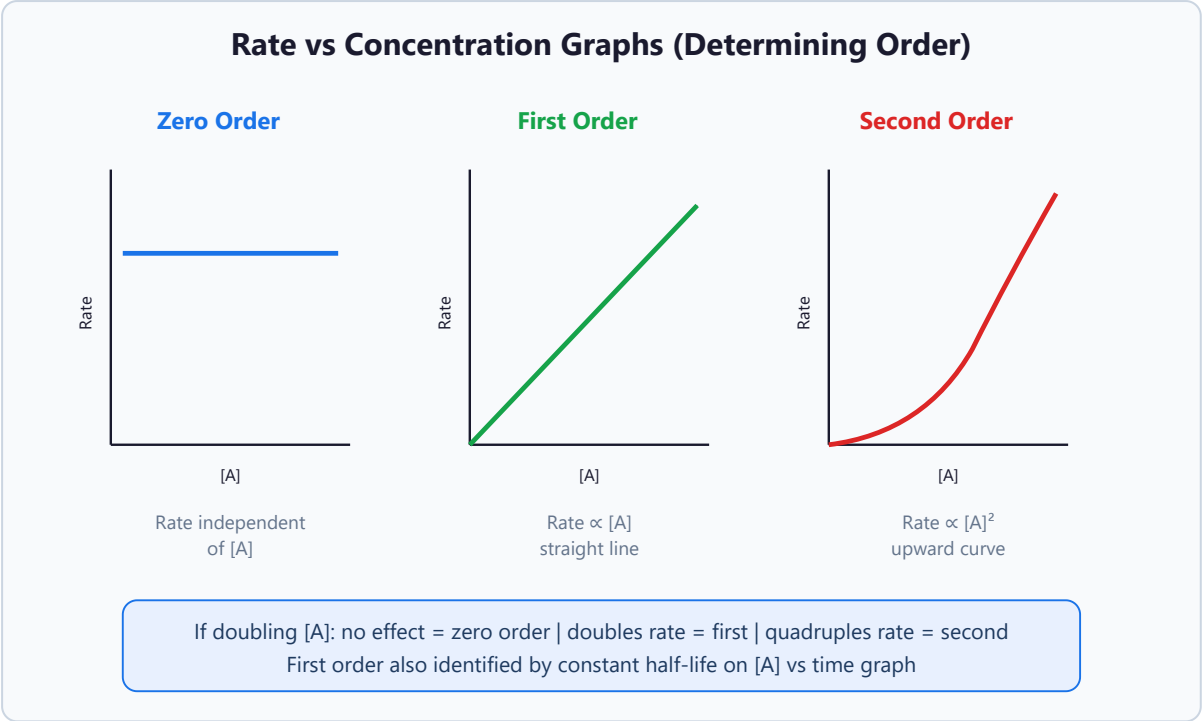


Determining Orders from Experimental Data

Using the **initial rates method**: compare experiments where only one concentration changes at a time while all others are held constant.

- If doubling [A] has **no effect** on rate → zero order in A
- If doubling [A] **doubles** the rate → first order in A
- If doubling [A] **quadruples** the rate → second order in A

More generally: if [A] is multiplied by factor x, and rate changes by factor x^n, then the order with respect to A is n.

Worked Example — Determining the Rate Equation:

Experiment	[A] / mol dm^{-3}	[B] / mol dm^{-3}	Initial rate / mol dm^{-3} s^{-1}
1	0.10	0.10	2.0×10^{-4}
2	0.20	0.10	8.0×10^{-4}

3	0.10	0.20	2.0×10^{-4}

Order w.r.t. A: Compare Exp 1 and 2 ([B] constant). [A] doubles, rate quadruples (2.0 → 8.0, ×4). Order = **2**

Order w.r.t. B: Compare Exp 1 and 3 ([A] constant). [B] doubles, rate unchanged. Order = **0**

Rate equation: Rate = $k[A]^2$

Calculate k: k = Rate / $[A]^2$ = 2.0×10^{-4} / $(0.10)^2$ = **0.020 dm^3 mol^{-1} s^{-1}**

The Rate Constant k

Rate constant (k): The proportionality constant in the rate equation. Its value is independent of concentration but **increases with temperature** (an increase in temperature always increases k). A larger k means a faster reaction at that temperature.

Units of k depend on the overall order:

Overall Order	Units of k
0	mol dm^{-3} s^{-1}
1	s^{-1}
2	dm^3 mol^{-1} s^{-1}

The Arrhenius Equation

$$k = Ae^{-E_a/RT}$$

Where A = the pre-exponential (Arrhenius) factor (related to collision frequency and orientation), E_a = activation energy (J mol^{-1}), R = 8.314 J K^{-1} mol^{-1}, T = temperature (K). Taking natural logarithms gives a linear form:

$$\ln k = \ln A - E_a/RT$$

A plot of **ln k vs 1/T** gives a straight line with gradient = **$-E_a/R$** and y-intercept = ln A. This allows E_a to be determined graphically: E_a = −gradient × R.

Exam Tip: When calculating E_a from the Arrhenius equation, ensure T is in kelvin and E_a comes out in J mol^{-1} (not kJ mol^{-1}). Divide by 1000 at the end if the question asks for kJ. When reading the gradient from a graph, use a large triangle and show coordinates clearly. The gradient will be negative (downward slope); multiply by −R to get a positive E_a.

Rate-Determining Step and Reaction Mechanisms

Rate-determining step (RDS): The slowest step in a multi-step reaction mechanism. It acts as a bottleneck, controlling the overall rate. The species that appear in the rate equation are those involved in or before the rate-determining step.

The rate equation provides evidence for the mechanism. If the rate equation is Rate = k[A][B], then both A and B are involved in (or before) the RDS. If a reactant does *not* appear in the rate equation, it is involved in a faster step *after* the RDS.

Worked Example — Linking Rate Equation to Mechanism:

The reaction between propanone and iodine in acid solution has the rate equation: Rate = $k[CH_3COCH_3][H^+]$

This tells us: (1) the reaction is first order in propanone and first order in H^+, (2) iodine does NOT appear in the rate equation so it is not involved in the RDS, (3) the RDS involves one molecule of propanone and one H^+ ion, (4) iodine must react in a fast step after the RDS.

5.2 How Far? — Equilibrium Constants

K_c Calculations

Use an **ICE (Initial, Change, Equilibrium)** table to determine equilibrium concentrations, then substitute into the K_c expression. Remember to convert amounts (moles) to concentrations (mol dm^{-3}) by dividing by the volume of the container in dm^3.

K_p — Equilibrium Constant in Terms of Partial Pressures

Partial pressure: The pressure exerted by an individual gas in a mixture. It is the contribution that gas makes to the total pressure. It equals the mole fraction of that gas multiplied by the total pressure.

$$p_A = x_A \times P_{total} \quad \text{where } x_A = \text{moles of A / total moles of gas}$$

For the reaction: aA(g) + bB(g) ⇌ cC(g) + dD(g)

$$K_p = (p_C)^c(p_D)^d / (p_A)^a(p_B)^b$$

K_p, like K_c, is only affected by temperature. Include units for K_p (typically in terms of the pressure unit used, e.g. Pa, kPa, or atm). Only gaseous species are included in K_p expressions.

Worked Example — Calculating K_p:

$N_2O_4(g) \rightleftharpoons 2NO_2(g)$. At equilibrium, the mole fractions are: $x(N_2O_4) = 0.30$, $x(NO_2) = 0.70$. Total pressure = 100 kPa.

$p(N_2O_4) = 0.30 \times 100 = 30$ kPa
$p(NO_2) = 0.70 \times 100 = 70$ kPa

$K_p = p(NO_2)^2 / p(N_2O_4) = 70^2 / 30 = 4900 / 30 =$ **163 kPa**

5.3 Acids, Bases and Buffers

Bronsted-Lowry Theory

Bronsted-Lowry acid: A proton (H^+) donor.

Bronsted-Lowry base: A proton (H^+) acceptor.

A **conjugate acid-base pair** differs by one proton.

pH Calculations

$$pH = -\log_{10}[H^+] \qquad [H^+] = 10^{-pH}$$

Strong acids fully dissociate, so $[H^+]$ = concentration of a monobasic acid. For dibasic strong acids like H_2SO_4, $[H^+]$ = 2 × concentration.

Strong bases: Use K_w to find $[H^+]$:

$$K_w = [H^+][OH^-] = 1.00 \times 10^{-14}\ mol^2\ dm^{-6} \text{ at } 25\ °C$$

Worked Example — pH of a Strong Base:

Calculate the pH of 0.050 mol dm^{-3} NaOH at 25 °C.

NaOH fully dissociates: $[OH^-]$ = 0.050 mol dm^{-3}

$[H^+] = K_w / [OH^-] = 1.00 \times 10^{-14} / 0.050 = 2.00 \times 10^{-13}$ mol dm^{-3}

$pH = -\log(2.00 \times 10^{-13})$ = **12.7**

Weak Acids and K_a

Weak acid: An acid that only partially dissociates in aqueous solution, establishing an equilibrium. The vast majority of molecules remain undissociated. Examples: CH_3COOH, H_2CO_3, HF.

$$K_a = [H^+][A^-] / [HA]$$

Two key assumptions for weak acid pH calculations (state these in exams for full marks):

1. $[H^+] = [A^-]$ because they are produced in equal amounts from the dissociation and the contribution of H^+ from water is negligible.
2. $[HA]_{equilibrium} \approx [HA]_{initial}$ because dissociation is so small that the change in [HA] is negligible.

$$K_a \approx [H^+]^2 / [HA]_{initial} \quad \text{so} \quad [H^+] = \sqrt{(K_a \times [HA])}$$

Exam Tip: Always state the two assumptions when performing weak acid calculations. Examiners frequently award marks specifically for these. Also remember: $pK_a = -\log K_a$, and a smaller K_a (or larger pK_a) means a weaker acid.

Buffer Solutions

Buffer solution: A solution that resists changes in pH when small amounts of acid or base are added. An **acidic buffer** (pH < 7) consists of a weak acid and its conjugate base (salt), e.g. CH_3COOH + CH_3COONa. A **basic buffer** (pH > 7) consists of a weak base and its conjugate acid.

How an acidic buffer works:

- **When H^+ is added:** The conjugate base (A^-, present in large excess from the salt) reacts with the added H^+: $A^- + H^+ \rightarrow HA$. This removes the excess H^+, so pH barely changes.
- **When OH^- is added:** The weak acid (HA, present in large reserve) reacts with OH^-: $HA + OH^- \rightarrow A^- + H_2O$. This removes the excess OH^-, so pH barely changes.

The buffer works because it contains large reserves of *both* HA and A^-, which can neutralise added acid or base respectively.

To calculate the pH of a buffer:

$$[H^+] = K_a \times [HA] / [A^-]$$

Titration Curves

Titration Type	pH at Equivalence Point	Suitable Indicator	Shape of Buffer Region
Strong acid + Strong base	7	Either methyl orange or phenolphthalein	Sharp vertical section; no buffer region
Strong acid + Weak base	< 7	Methyl orange (range 3.1–4.4)	Buffer region after equivalence
Weak acid + Strong base	> 7	Phenolphthalein (range 8.3–10.0)	Buffer region before equivalence
Weak acid + Weak base	~7	No suitable indicator; use a pH meter	Very gradual change; no sharp equivalence

The **half-equivalence point** (or half-neutralisation point) is where exactly half the acid has been neutralised. At this point, $[HA] = [A^-]$, so $[H^+] = K_a$, meaning **$pH = pK_a$**. This is the point of maximum buffering capacity and can be used to determine K_a from a titration curve.

Exam Tip: An indicator is suitable if its colour change range ($pK_{In} \pm 1$) falls within the steep vertical section of the titration curve. For weak acid + weak base titrations, the curve has no steep section, so no indicator works — use a pH meter instead.

5.4 Enthalpy, Entropy and Free Energy

Lattice Enthalpy

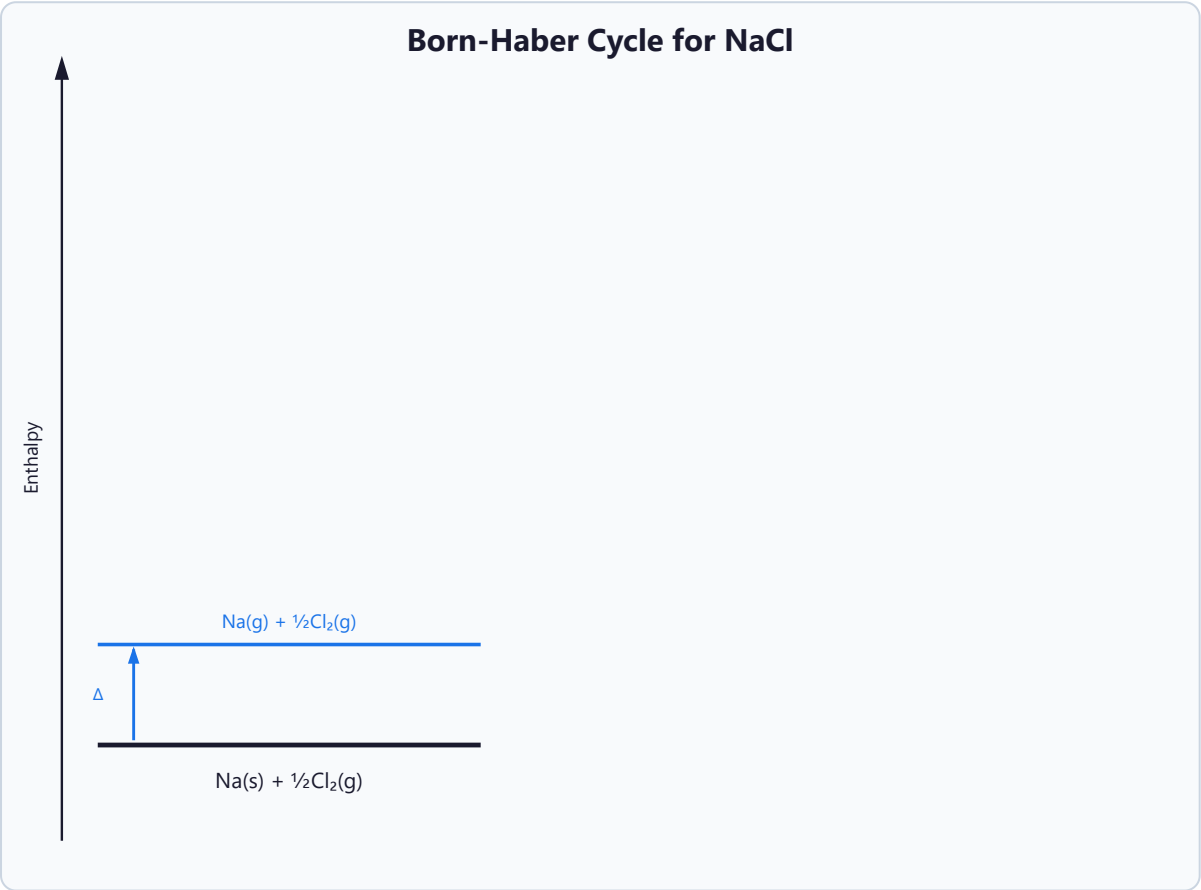
Lattice enthalpy of formation: The enthalpy change when one mole of an ionic lattice is formed from its gaseous ions under standard conditions. It is always **exothermic** (negative). Alternatively defined as lattice enthalpy of dissociation (endothermic, positive) — check which definition the exam board uses.

Lattice enthalpy becomes more exothermic (stronger lattice) with:

- **Smaller ionic radii** — ions are closer together, so electrostatic attraction is stronger
- **Higher ionic charges** — greater charge means stronger electrostatic attraction

For example, MgO has a much more exothermic lattice enthalpy than NaCl because Mg^{2+} and O^{2-} have higher charges and smaller radii than Na^+ and Cl^-.

Born-Haber Cycles



$_{at}H$ (Na) Na(g) + Cl(g) $\Delta_{at}H$ (Cl) $Na^+(g) + e^- + Cl(g)$ IE_1 (Na) $Na^+(g) + Cl^-(g)$ EA_1 (Cl) NaCl(s) $\Delta_{latt}H°$ (lattice enthalpy) $\Delta_fH°$ (NaCl) Steps (all endothermic upward) 1. Atomise metal: Na(s) → Na(g) 2. Atomise non-metal: $½Cl_2$ → Cl(g) 3. Ionise metal: Na(g) → $Na^+(g)$ 4. Electron affinity: Cl(g) → $Cl^-(g)$ 5. Form lattice (exothermic) Lattice H more exothermic with smaller ions and higher charges

A Born-Haber cycle applies Hess's Law to calculate the lattice enthalpy of an ionic compound when it cannot be measured directly. The cycle includes: enthalpies of atomisation, ionisation energies (for the metal), electron affinities (for the non-metal), and enthalpy of formation. The lattice enthalpy is the unknown that completes the cycle.

Enthalpy of Hydration and Solution

$$\Delta_{sol}H = -\Delta_{latt}H + \Sigma\Delta_{hyd}H$$

For a salt to dissolve, the energy released by hydration of the gaseous ions ($\Delta_{hyd}H$, always exothermic) must compensate for the energy needed to break apart the lattice ($-\Delta_{latt}H$, endothermic). Enthalpy of hydration is more exothermic for ions with a **higher charge density** (smaller radius or higher charge), because they attract water molecules more strongly.

Entropy

Entropy (S): A quantitative measure of the disorder (or number of possible arrangements of particles) in a system. Units: J K^{-1} mol^{-1}. Higher entropy = more disorder = more ways to arrange the particles.

Entropy increases when:

- A solid melts to a liquid, or a liquid boils to a gas (gas >> liquid > solid)
- A solid dissolves in a solvent (particles become dispersed)
- The number of gaseous moles increases in a reaction
- Temperature increases (particles have more energy and more possible arrangements)

$$\Delta S_{system} = \Sigma S(\text{products}) - \Sigma S(\text{reactants})$$

Gibbs Free Energy

$$\Delta G = \Delta H - T\Delta S$$

A reaction is **thermodynamically feasible** (spontaneous) when $\Delta G < 0$. However, even if ΔG is negative, the reaction may still not occur at an observable rate if the activation energy is too high (kinetic barrier). Feasibility tells us whether a reaction *can* happen, not whether it *will* happen quickly.

ΔH	ΔS	Feasibility (ΔG < 0?)
Negative (exo)	Positive (more disorder)	**Feasible at all temperatures** (ΔG always negative)

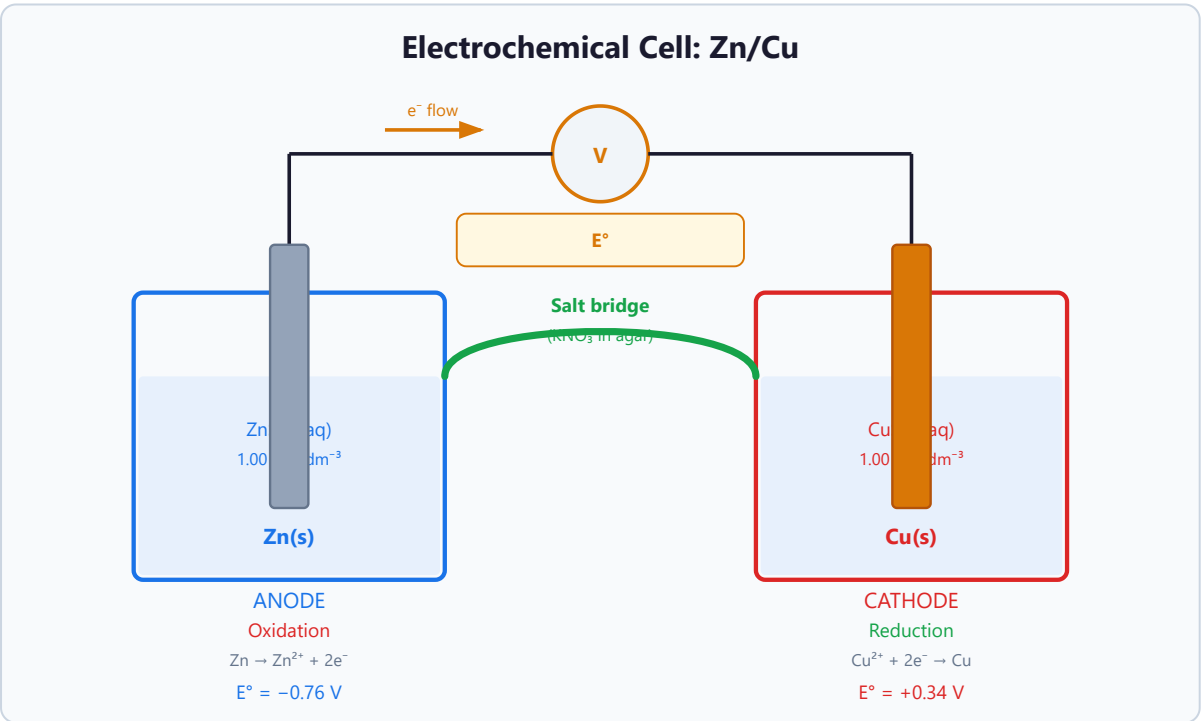
Negative (exo)	Negative (less disorder)	Feasible at **low temperatures** only (when $\lvert T\Delta S\rvert < \lvert\Delta H\rvert$)
Positive (endo)	Positive (more disorder)	Feasible at **high temperatures** only (when $T\Delta S > \Delta H$)
Positive (endo)	Negative (less disorder)	**Never feasible** (ΔG always positive)

Exam Tip: When using $\Delta G = \Delta H - T\Delta S$, ensure consistent units. If ΔH is in kJ mol^{-1} and ΔS is in J K^{-1} mol^{-1}, convert ΔS to kJ (divide by 1000) or ΔH to J (multiply by 1000) before calculating. Temperature must be in kelvin. To find the minimum temperature for feasibility, set $\Delta G = 0$ and solve: $T = \Delta H / \Delta S$.

5.5 Redox and Electrode Potentials

Standard Electrode Potential

Standard electrode potential (E°): The EMF (voltage) of a half-cell measured under standard conditions (298 K, 100 kPa, 1.00 mol dm^{-3} solutions) relative to the **standard hydrogen electrode (SHE)**, which is assigned a value of exactly 0.00 V.



$_{cell} = +0.34 - (-0.76)$

$= +1.10$ V

Electrochemical Cells

$$E°_{cell} = E°_{cathode} - E°_{anode} = E°(\text{more positive}) - E°(\text{more negative})$$

A **positive $E°_{cell}$** indicates the reaction is thermodynamically feasible under standard conditions. The more positive half-cell undergoes reduction (cathode); the more negative half-cell undergoes oxidation (anode). Electrons flow from anode to cathode through the external circuit. The salt bridge completes the circuit by allowing ions to flow between the half-cells, maintaining electrical neutrality.

Limitations of E° predictions: A positive $E°_{cell}$ does not guarantee the reaction will actually occur at an observable rate because: (1) the activation energy may be too high (kinetic barrier), (2) conditions may not be standard, (3) E° values apply to standard conditions only — non-standard concentrations, temperatures, or pressures will shift the value.

Fuel Cells

> **Fuel cell:** An electrochemical cell that converts the chemical energy of a fuel (e.g. hydrogen) directly into electrical energy, as long as fuel and oxidant are continuously supplied. Unlike batteries, fuel cells do not run down or need recharging.

In a hydrogen fuel cell:

- **Anode:** $H_2(g) \rightarrow 2H^+(aq) + 2e^-$ (oxidation)
- **Cathode:** $O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(l)$ (reduction)
- **Overall:** $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$

Advantages: only product is water (zero emissions at point of use), higher theoretical efficiency than combustion engines (not limited by Carnot cycle), quiet operation.

Challenges: hydrogen production is currently energy-intensive (often from steam reforming of methane, which produces CO_2), storage requires high pressure (700 bar) or cryogenic temperatures, distribution infrastructure is limited, platinum catalyst is expensive and scarce.

5.6 Transition Elements

Transition element: A d-block element that forms at least one stable ion with an **incomplete d sub-shell**. This definition excludes scandium (Sc^{3+} is [Ar] $3d^0$ — empty d shell) and zinc (Zn^{2+} is [Ar] $3d^{10}$ — full d shell).

Properties of Transition Elements

- **Variable oxidation states:** Due to the similar energies of the 3d and 4s sub-shells, transition metals can lose different numbers of electrons to form ions with different charges (e.g. Fe^{2+} and Fe^{3+}, Cu^{+} and Cu^{2+}, Mn^{2+} to Mn^{7+}). This allows them to participate in redox reactions.
- **Formation of coloured complexes:** Caused by **d-d transitions**. When ligands surround a transition metal ion, the d orbitals split into two energy levels. Electrons absorb visible light of a specific wavelength and transition between these levels. The colour observed is **complementary** to the colour absorbed. If the d sub-shell is empty (d^0) or full (d^{10}), no d-d transition is possible and the compound is white or colourless.
- **Catalytic activity:** Transition metals act as catalysts due to their ability to change oxidation state (providing alternative pathways) and provide a surface for adsorption of reactants. Examples: Fe in the Haber process, V_2O_5 in the Contact process, MnO_2 in decomposition of H_2O_2, Ni in hydrogenation.
- **Formation of complex ions** with ligands (see below).

Complex Ions

Complex ion: A central metal ion surrounded by ligands. A **ligand** is a molecule or ion that donates a **lone pair** of electrons to the central metal ion to form a **dative (coordinate) bond**. The **coordination number** is the total number of dative bonds to the central metal ion.

Ligand	Type	Lone Pairs Donated	Example Complex
H_2O	Monodentate	1	$[Cu(H_2O)_6]^{2+}$ (pale blue)
NH_3	Monodentate	1	$[Cu(NH_3)_4(H_2O)_2]^{2+}$ (deep blue)
Cl^-	Monodentate	1	$[CuCl_4]^{2-}$ (yellow-green)

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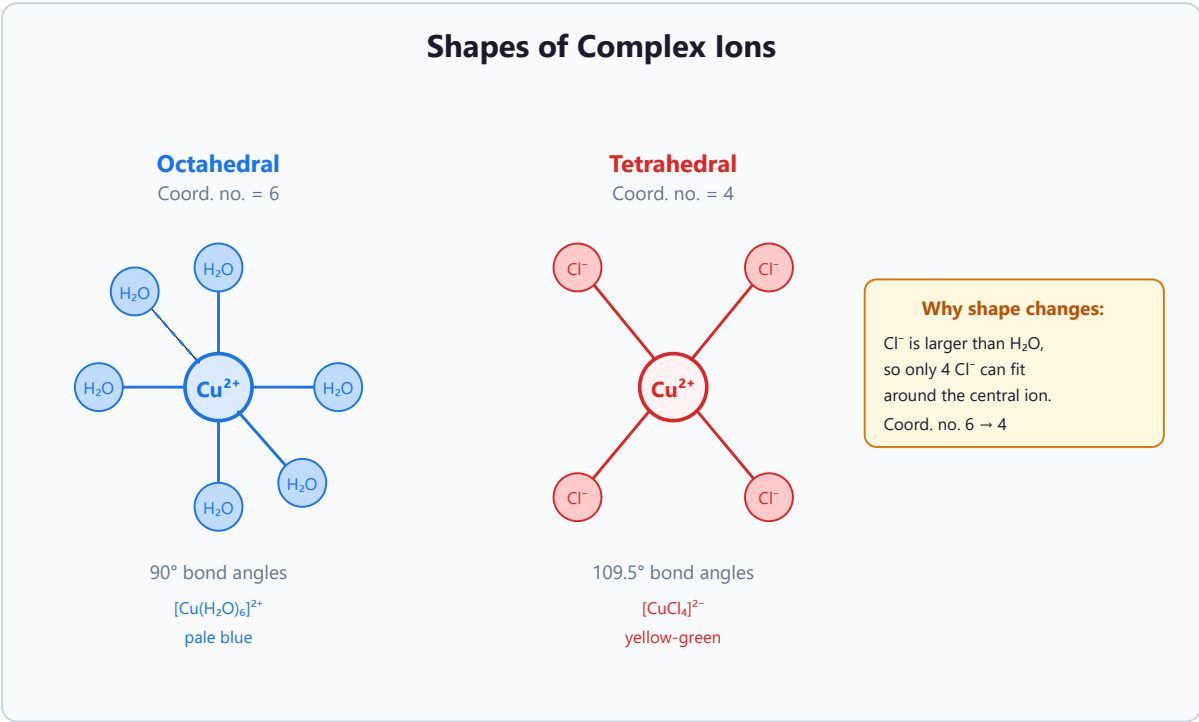
$NH_2CH_2CH_2NH_2$ (en)	Bidentate	2	$[Ni(en)_3]^{2+}$
$EDTA^{4-}$	Hexadentate	6	$[Fe(EDTA)]^-$

Monodentate: A ligand that donates one lone pair (forms one dative bond). **Bidentate:** Donates two lone pairs from two different atoms. **Hexadentate (polydentate):** Donates six lone pairs, e.g. EDTA wraps around the central ion, forming 6 bonds.



Ligand Substitution

Ligands can be exchanged for other ligands. This may change the colour, coordination number, and shape of the complex:

- $\mathbf{[Cu(H_2O)_6]^{2+}}$ (pale blue, octahedral) + $4Cl^-$ ⇌ $\mathbf{[CuCl_4]^{2-}}$ (yellow-green, tetrahedral) + $6H_2O$ — coordination number changes from 6 to 4 because Cl^- is larger than H_2O
- $\mathbf{[Cu(H_2O)_6]^{2+}}$ (pale blue) + $4NH_3$ → $\mathbf{[Cu(NH_3)_4(H_2O)_2]^{2+}}$ (deep blue) + $4H_2O$ — partial substitution only (4 of 6 water molecules replaced), coordination number stays at 6, shape stays octahedral

The **chelate effect**: complexes with polydentate ligands (e.g. EDTA, en) are more stable than those with equivalent monodentate ligands because the substitution reaction has a positive ΔS (more particles are

released than consumed), making ΔG more negative.

$_g$ t_{2g} ΔE hν Why complexes are coloured: 1. Ligands cause d orbitals to split into two energy levels 2. Electrons absorb visible light (ΔE = hν) for d-d transitions 3. Colour seen = complementary to colour absorbed

Colour of Complexes

The colour depends on: (1) the identity of the central metal ion, (2) its oxidation state, (3) the identity of the ligands (different ligands cause different degrees of splitting), and (4) the coordination number/geometry. Any change in these factors can change the colour.

Exam Tip: When explaining why transition metal complexes are coloured, you must mention all three points: (1) d orbital splitting by ligands, (2) d-d electron transitions absorbing specific wavelengths of visible light, (3) the transmitted/reflected colour is complementary to the absorbed colour. If the d sub-shell is empty (d^0, e.g. Sc^{3+}) or full (d^{10}, e.g. Zn^{2+}, Cu^+), no d-d transition is possible and the compound is white or colourless.

Precipitation Reactions

Adding NaOH(aq) or NH_3(aq) to solutions of transition metal ions produces coloured precipitates of the metal hydroxide. Some of these dissolve in excess reagent:

Metal Ion	Colour of Hydroxide Precipitate	With Excess NaOH	With Excess NH_3
Cu^{2+}	Blue	No change (insoluble)	Dissolves: deep blue $[Cu(NH_3)_4(H_2O)_2]^{2+}$
Fe^{2+}	Green (darkens on standing as Fe^{2+} oxidises to Fe^{3+})	No change	No change
Fe^{3+}	Brown/rust (red-brown)	No change	No change
Cr^{3+}	Green	Dissolves: dark green $[Cr(OH)_6]^{3-}$ (amphoteric)	Dissolves: purple $[Cr(NH_3)_6]^{3+}$
Mn^{2+}	Pale brown/cream (darkens in air)	No change	No change